Sebastian Victor Dawber

W1610495

DES3B portfolio

Hemp hurds (3)



Lime binder (1)



Water (1)



Mix: hurd, binder .



Mix: Add water.



Mix: stir until even



Create mould box



Pour into box



Leave it to set



Final outcome



Final outcome



Final outcome



Design startegy: Concept.....3

Design startegy: Hemp as a plant.....4

Design startegy: Precedents.....5

Fabrication process and technologies: Process.....6

Geometry and digital form finding: Material testing.....7

Geometry and digital form finding: Form finding.....8

Performative testing: Unit testing.....9

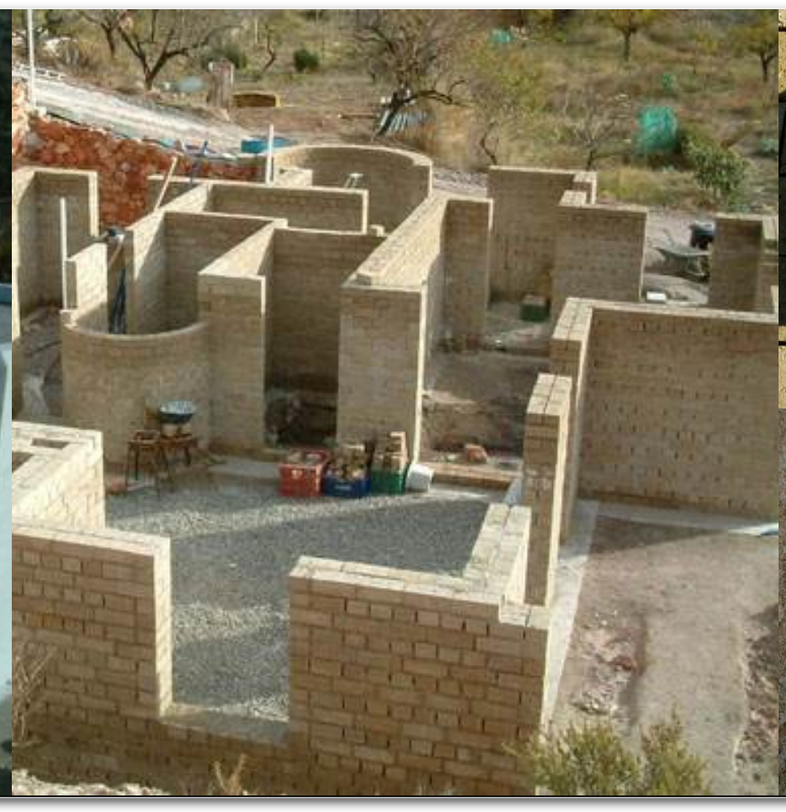
Performative testing: Enviromental dirgams.....10

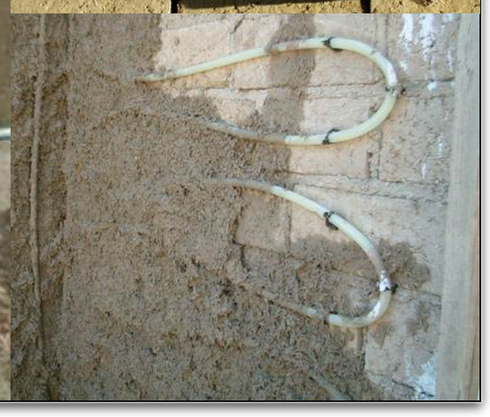
Performative testing: Detail section start of catalogue.....11

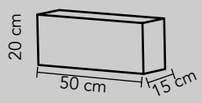
Assembly startegy and constrcution system: Catalogue.....12

Assembly startegy and constrcution system: Assembly.....13

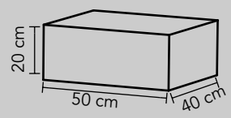
single-family house 2005

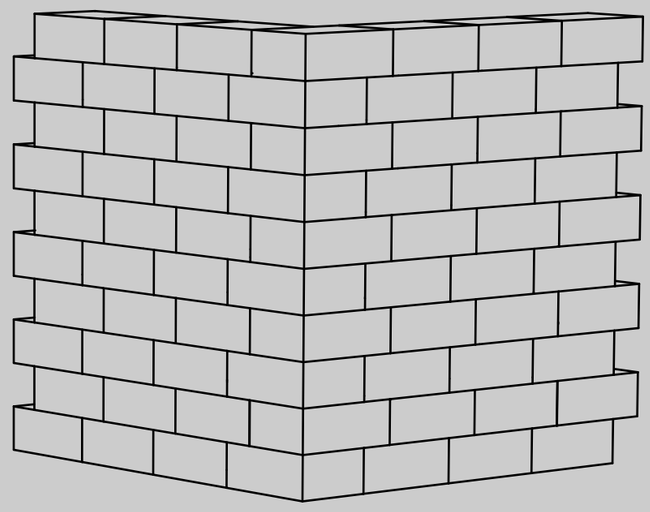




Thickness: 15cm	
Density (kg m/3)	330
Thermal conductivity (W/(mK)	0.07
Thermal transmittance U-vlaue (W/(m2K)	0.53
Resistance to water vapour (m)	4.50
Heat capacity (K/kgK)	1870
Sound absorption	0.8
Sound reduction	36.55



Thickness: 40cm	
Density (kg m/3)	330
Thermal conductivity (W/(mK)	0.07
Thermal transmittance U-vlaue (W/(m2K)	0.17
Resistance to water vapour (m)	4.50
Heat capacity (K/kgK)	1870
Sound absorption	0.8
Sound reduction	42.96



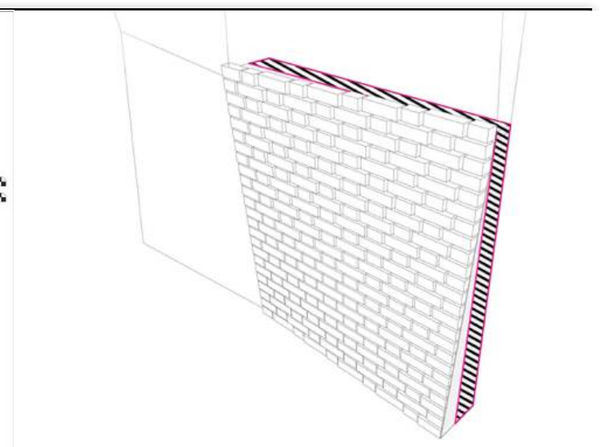
Cannabric, hempbricks. Compressed earth brick with hemp for insulating, loadbearing inside- and outside walls.

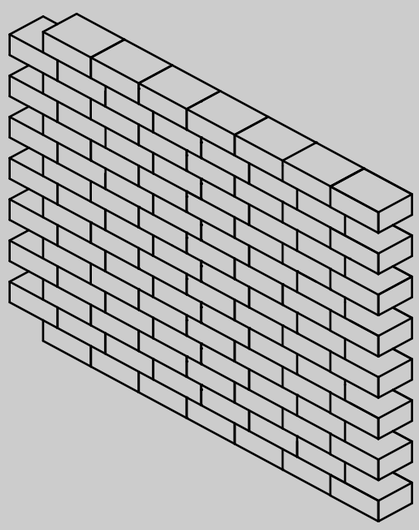
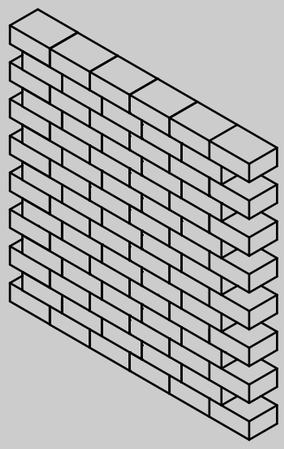
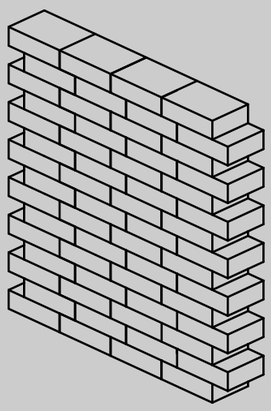
A nautural plant based material system using inudstrial hemp, when combined with a nautral binder and water forms into bulding compostiw material. It can used used as an extire wall with no extra insulation of internal parts needed, a vapour permeable lime render is used on the external side as the material is non-water resistant.

290 Mulberry / SHoP Architects





Due to struct building codes and the inability to build in the cofniend area SHoP archtiects had to create a system that could be assbmled while abding by the cityies strict bulding codes, due to the nature of the space conustrction work had to be minimizsed.

Making a literal interpretation of code written for classical ornamentation allowed us to project our enclosure over the property line at 10% intervals for every 100 square feet. Maximizing the amount of projected area, while minimizing the overall depth of the enclosure became key criteria for our design. When coupled with material properties and fabrication constraints, these criteria began to define an approach that was a contemporary reinterpretation of brick detailing. By customizing a standard precast brick panel system, we were able to achieve maximum effect at minimum cost. The building then becomes veiled by an textured wrapper around the street walls in contrast with the simplicity of the inner core.

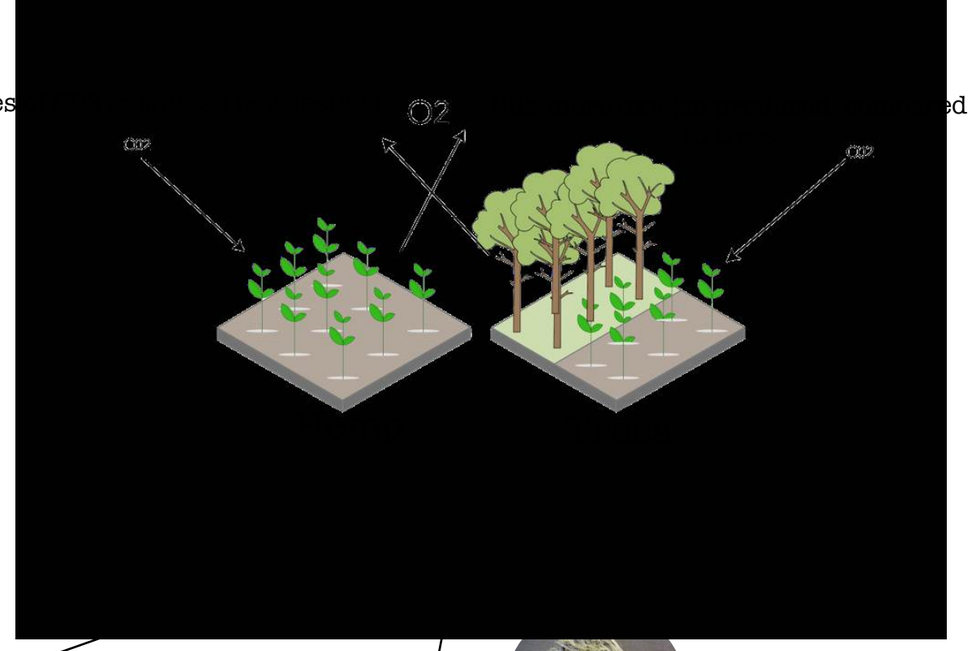
Materiality and physical study model

Hemp plant anatomy

Design startegy: Hemp as a plant

Hemp

15 tonnes





Seeds

Oil
cooking oil
diet supplement
beauty products
fuel
paint

Hemp nut
milk
bakery
granola
protien powder

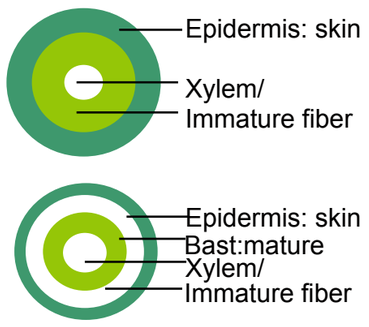
Stalk

Fiber
Texttiles
Insulation
Rope

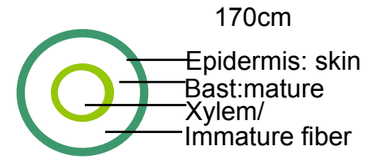
Hurds
Paper
Compost
Animla bedding
Fiber board
Hempcrete
Stucco
Mortar







5.5ft
170cm





Flowers
animal bedding
compost
medicine

Roots
medicine
compost

Hemp for hemp hurd extraction can be harvested in the vegatative stage



SEED

3 day



SEEDING

2 week



VEGETATIVE

4-8 weeks



FLOWERING

4-8 weeks



HARVEST

Varied

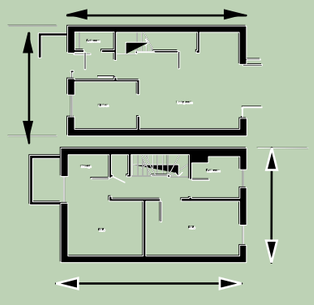
Process

Fabrication process and technologies: Process

1 Precedent study







Precedents were chosen to give the design a foundation of ideas, a base to start the design.

A precedent from DES3A was integrated as it gave dimensions of a house made from hemp, a link between the two projects but the housing links the project, the social housing project.

Canabric was used to see how hemp bricks can be assembled.

290 Mulberry allowed me to see how brick panvels could be assembled and constructed in a confined space

2 Material testing

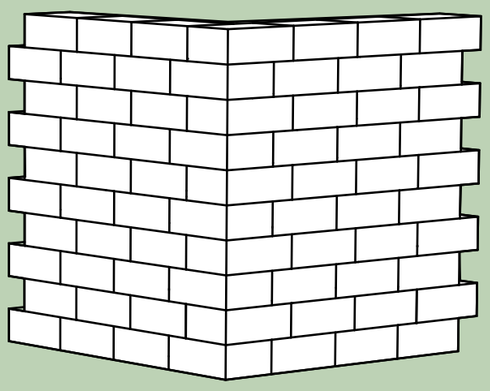




Material testing assessed the best ratio needed to create hemp bricks.

Hemp:Binder:Water – 2:1:1

3 Formation



+



Due to COVID-19 tests were limited and digital forms had to be constructed and chosen with little physical testing but more so online and digital research was used to find the final form. Using a stretcher brick bond and dimensions of 8mx5m.

4 Simulation



Once the form was found IES VE was used to get accurate environmental simulations. Allowing me to see that hempcrete works better with closed rooms rather than open spaces.

5 Construction



The simulation gave a final form with was created and edited on Rhinoceros 3D to construct windows and created the finished panels as well as form.

6 Layout



Graphic

Graphic was used to lay out and present the final work.

Materiality and physical study model

Material testing

Geometry and digital form finding: Material testing

"Ingredients" "Mix:Biner-Hurd-Water-Hurd" "Add Mix to box" "Test"

1.Binder (B)



2.Hemp hurds(H)



3.Water(W)



4.Mix: Binder-Hurds



5.Mix: Binder-Hurds-Water



7. Add rest of the hurds as you stir



8.Mix evenly



9.Pour mixture into mould box



10.Let it set for 48hrs



11.Create a ball



12.Poke a hole



17.If done correctly it sticks together

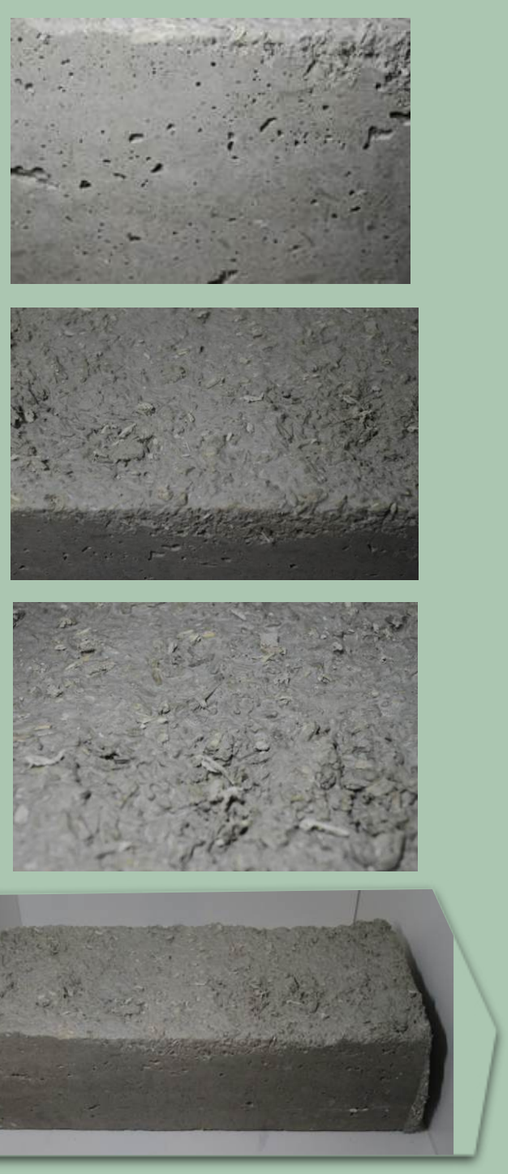


B:H:W- 1:3:1



To many hurds were added.
Bricks started breaking

B:H:W- 2:3:1



To mcuh binder was used

B:H:W- 1:2:1



-Uneven mix

B:H:W- 1:2:1

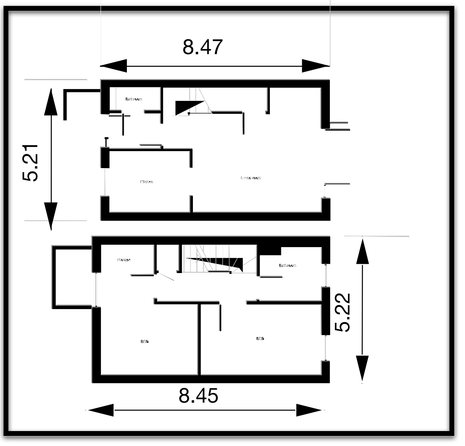


-Even mix

Perfect mix but mixing needs
to be done corretly

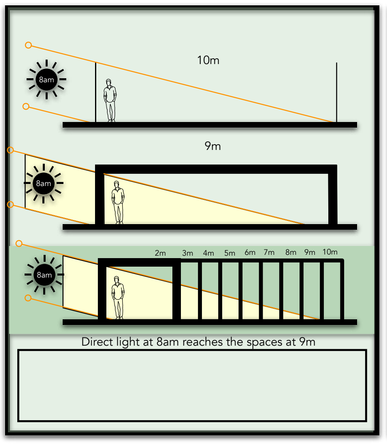
Adding a mix ratio of B:H:W- 1:2:1 leads to a successful hempcrete mixture

1 .Plans taken from Triangle Swindon, Glenn Howells architects. It gave dimensions for hempcrete housing, due to the house only having space for one person it was reduced to one floor with dimensions: 8.5 X 5.2m.



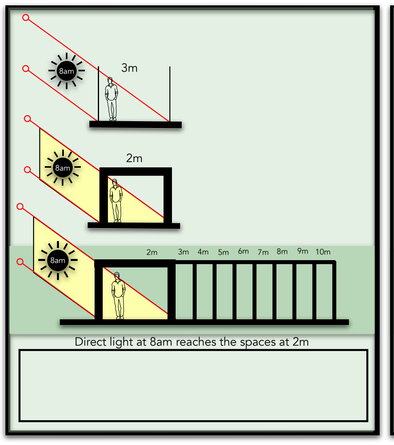
2. Sections were used to find the best length that will allow sun to enter the space at 8am, the time when residents wake up for work very morning.

22 March- Equinox solstice



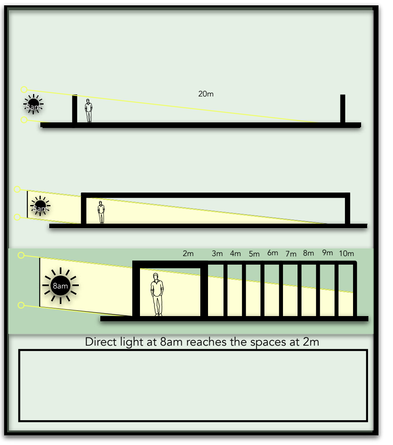
Direct light at 8am reaches the spaces at 9m

22 June- Summer solstice



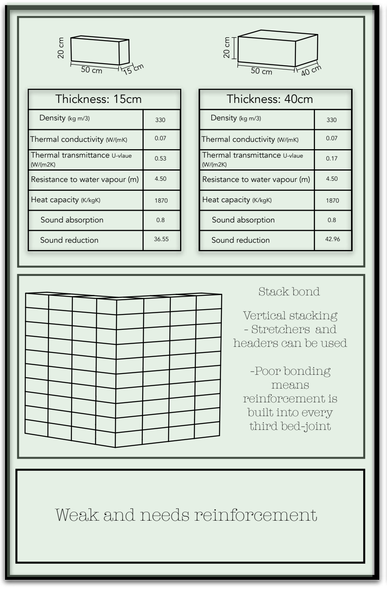
Direct light at 8am reaches the spaces at 2m

22 December- Wintersolstice



Direct light at 8am reaches the spaces at 2m

3. Three different wall bonding techqniues were exmained but due to COVID-19 testing was limited and a online research had to be used.



Thickness: 15cm	
Density (kg m^3)	330
Thermal conductivity (W/(mK)	0.07
Thermal transmittance U-vlaue (W/(m2K)	0.53
Resistance to water vapour (m)	4.50
Heat capacity (K/kgK)	1870
Sound absorption	0.8
Sound reduction	36.55

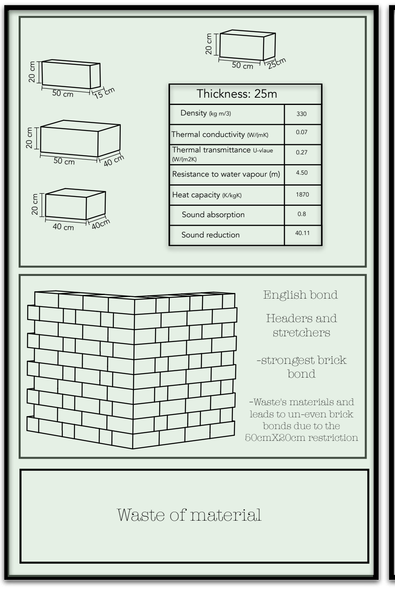
Thickness: 40cm	
Density (kg m^3)	330
Thermal conductivity (W/(mK)	0.07
Thermal transmittance U-vlaue (W/(m2K)	0.17
Resistance to water vapour (m)	4.50
Heat capacity (K/kgK)	1870
Sound absorption	0.8
Sound reduction	42.96

Stack bond

Vertical stacking - Stretchers and headers can be used

-Poor bonding means reinforcement is built into every third bed-joint

Weak and needs reinforcement



Thickness: 25m	
Density (kg m^3)	330
Thermal conductivity (W/(mK)	0.07
Thermal transmittance U-vlaue (W/(m2K)	0.27
Resistance to water vapour (m)	4.50
Heat capacity (K/kgK)	1870
Sound absorption	0.8
Sound reduction	40.11

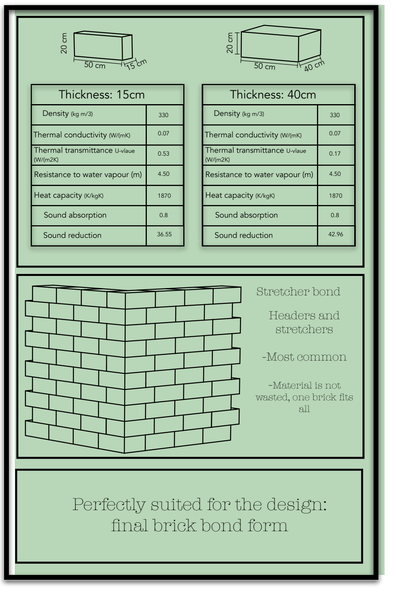
English bond

Headers and stretchers

-strongest brick bond

-Waste's materials and leads to un-even brick bonds due to the 50cmX20cm restriction

Waste of material



Thickness: 15cm	
Density (kg m^3)	330
Thermal conductivity (W/(mK)	0.07
Thermal transmittance U-vlaue (W/(m2K)	0.53
Resistance to water vapour (m)	4.50
Heat capacity (K/kgK)	1870
Sound absorption	0.8
Sound reduction	36.55

Thickness: 40cm	
Density (kg m^3)	330
Thermal conductivity (W/(mK)	0.07
Thermal transmittance U-vlaue (W/(m2K)	0.17
Resistance to water vapour (m)	4.50
Heat capacity (K/kgK)	1870
Sound absorption	0.8
Sound reduction	42.96

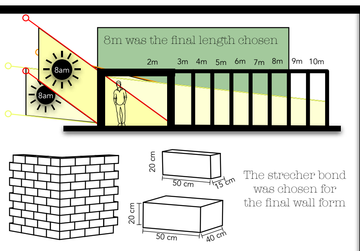
Stretcher bond

Headers and stretchers

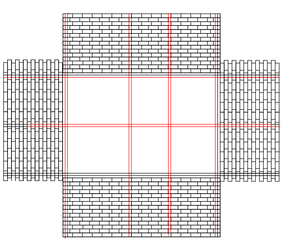
-Most common

-Material is not wasted, one brick fits all

Perfectly suited for the design: final brick bond form



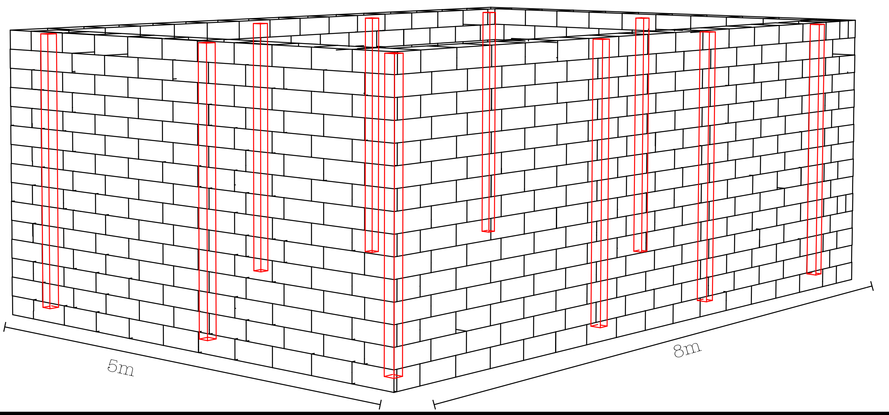
The strecher bond was chosen for the final wall form

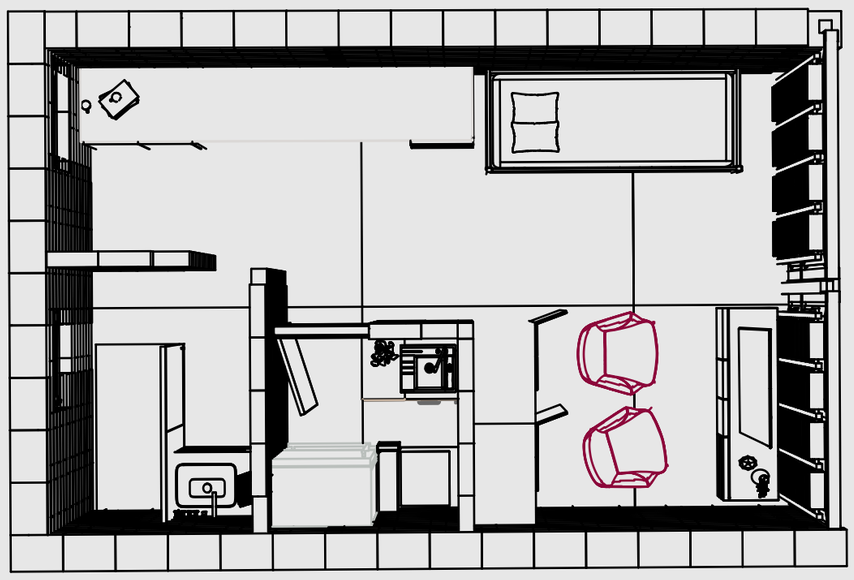


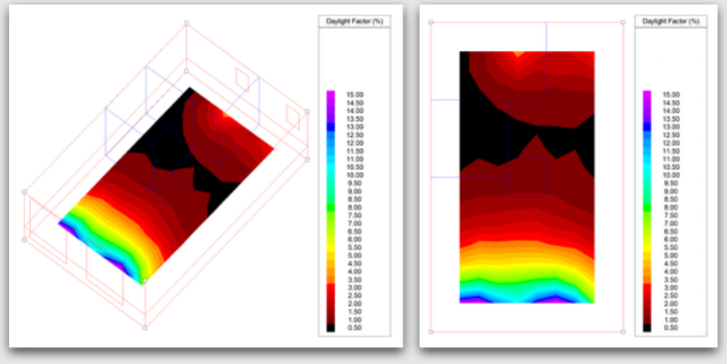
Final form

The final layout was developed using lines to place support pillars creating a catalouge of components

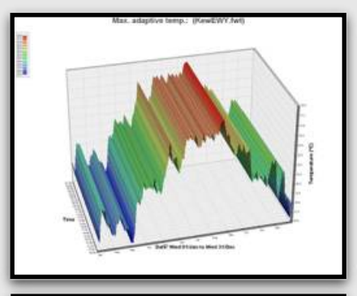
Using stretcher bond and a 5mx8m dimension similar to the Triangle Swindon, Glenn Howells architect precedent the social housing space was formed. Creating a single unit housing module that allows light to enter the main room at 8am every morning throughout the year., except for winter where it only enters at 9am

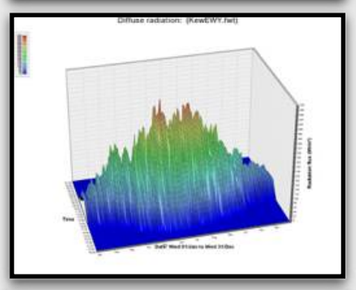


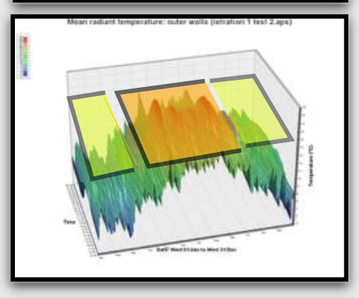


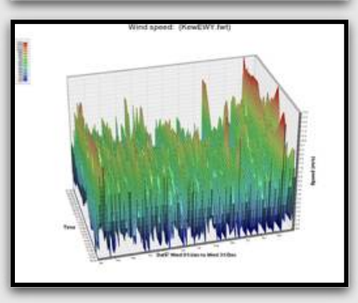


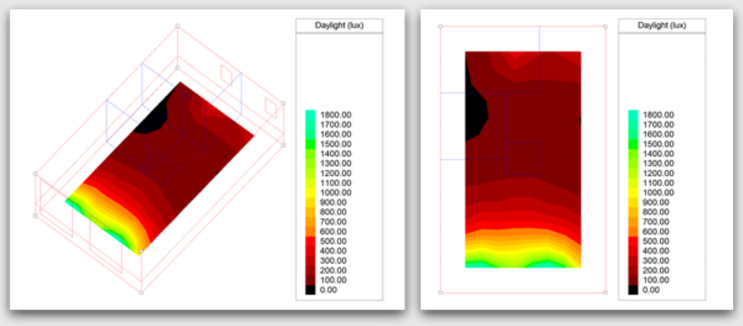
The high daylight factor and LUX level indicates the room is well lit but can be seen as too bright in areas. Light travels clearly through the entire space



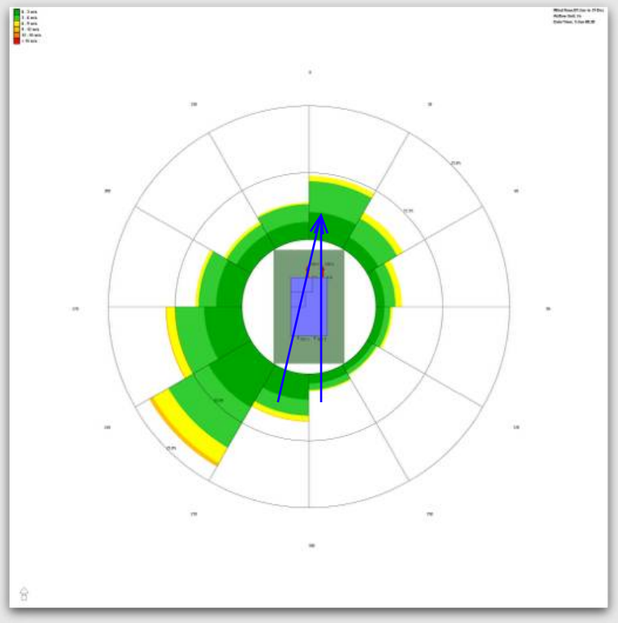


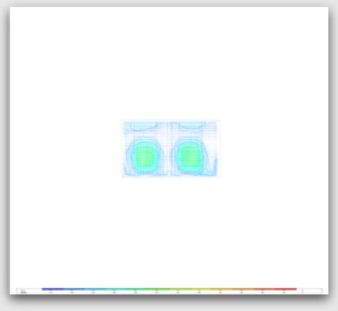


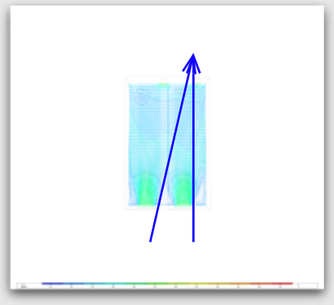


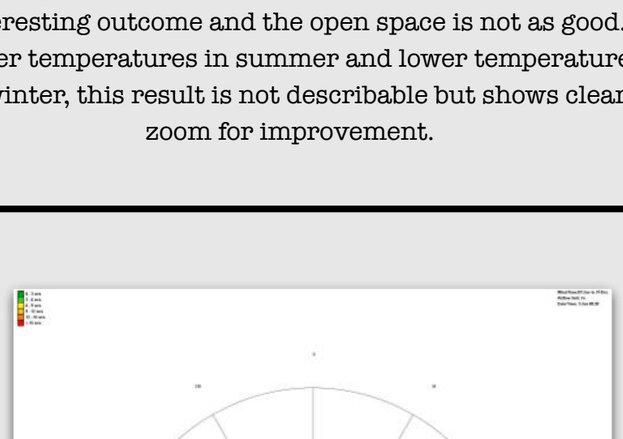


The first design uses an open floor plan to allow for a bigger space. The mean radiant temperature showed an interesting outcome and the open space is not as good. Higher temperatures in summer and lower temperatures in winter, this result is not describable but shows clear zoom for improvement.

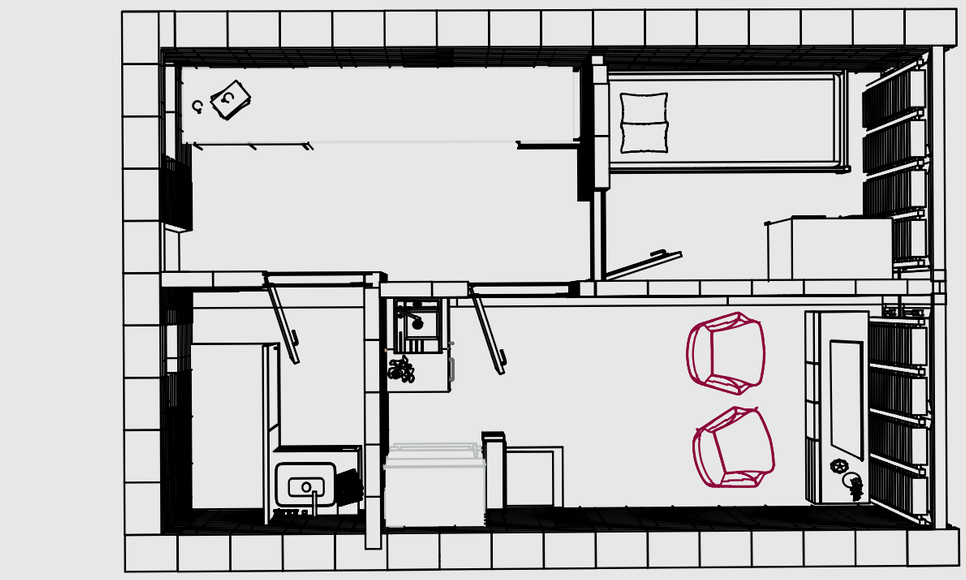


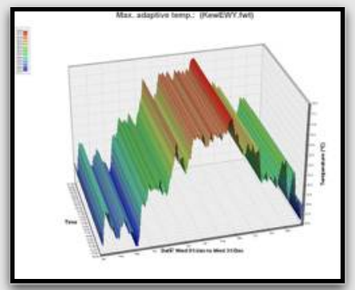


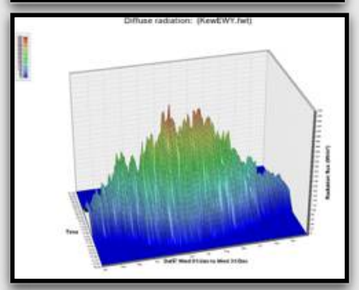


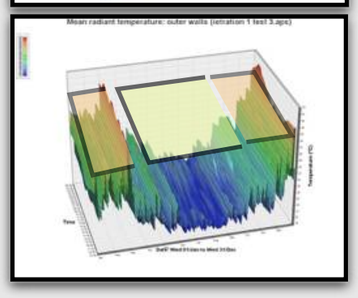


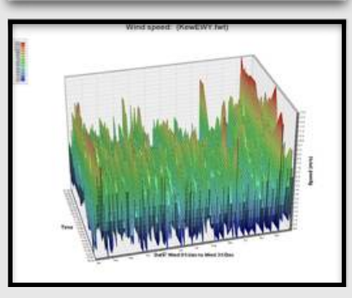
Adding windows on either side allows for cross ventilation. The open space had lower wind levels with wind being dispersed due to the open space nature. The wind travels from the floor and goes up, this may cause issues when walking around of working due to the wind currents coming from the floor.

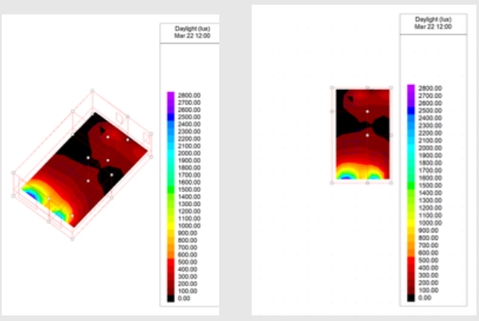




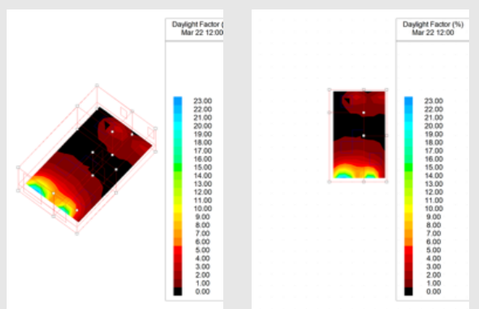




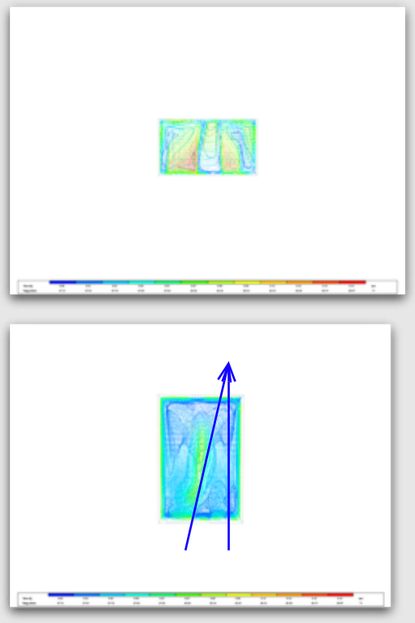


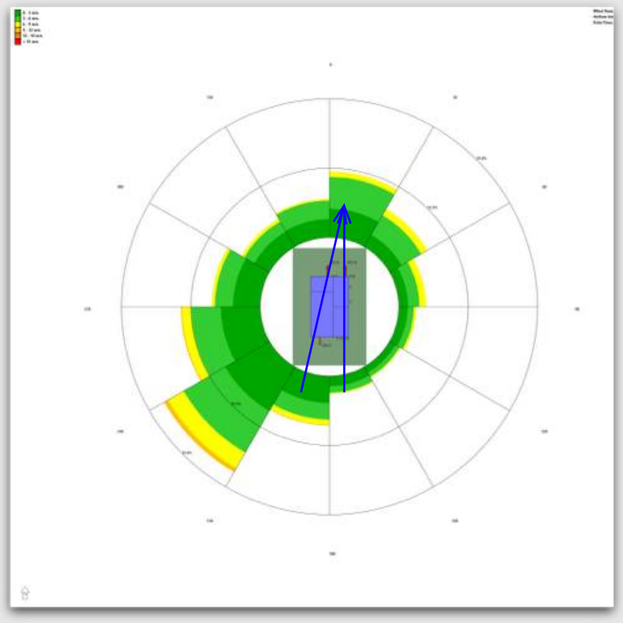


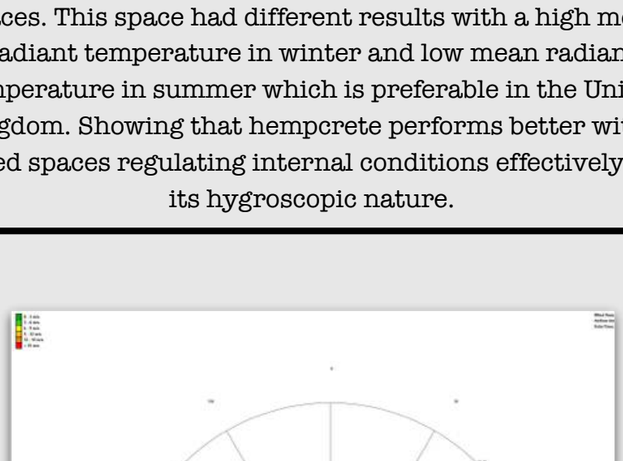
The daylight factor and LUX level indicates the room is well lit but has issues in areas that aren't close to the windows. Light travels through the space but doesn't perform as well as number 1.



The second design uses a closed floor plan to allow for a confined spaces. This space had different results with a high mean radiant temperature in winter and low mean radiant temperature in summer which is preferable in the United Kingdom. Showing that hempcrete performs better within confined spaces regulating internal conditions effectively due to its hygroscopic nature.



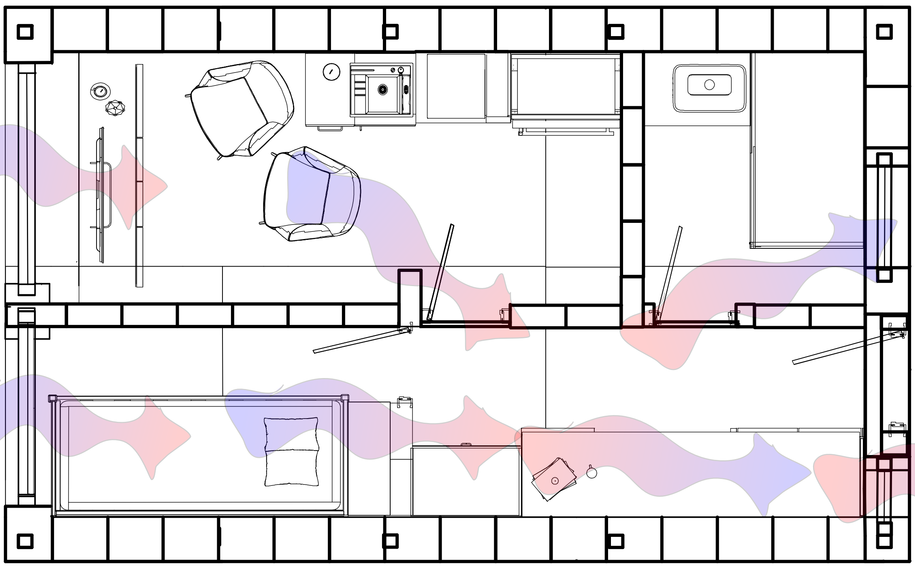


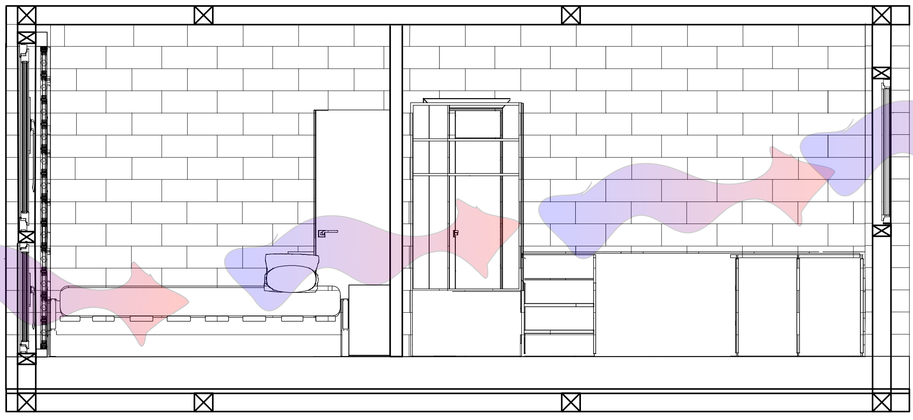


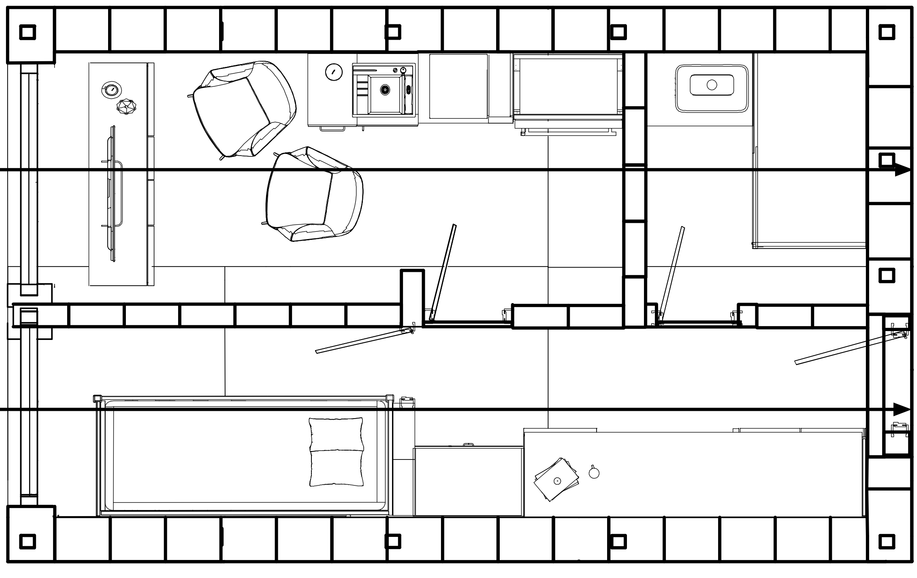
Two opening for cross ventilation lets air pass through the space freely. The closed space lets air travel above the inhabitants and has less effect on those using the space airing the unit and leaving with no issues from wind currents.

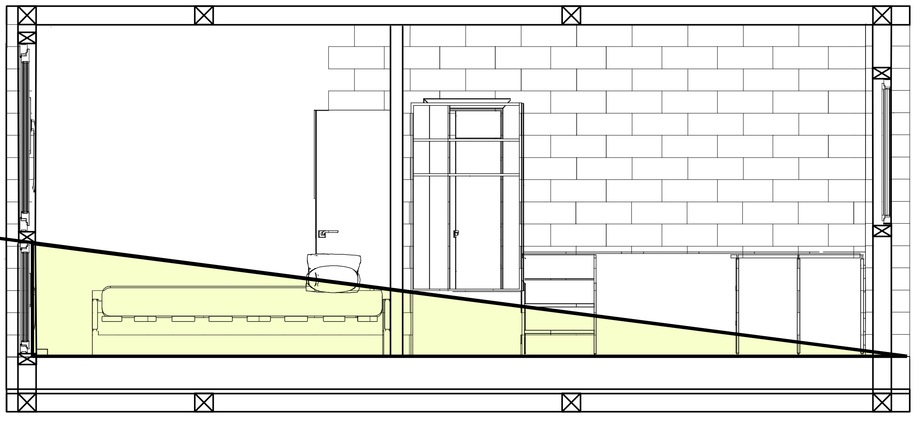
Plan and section 1:100; social housing

Performative testing: Enviromental dirgams

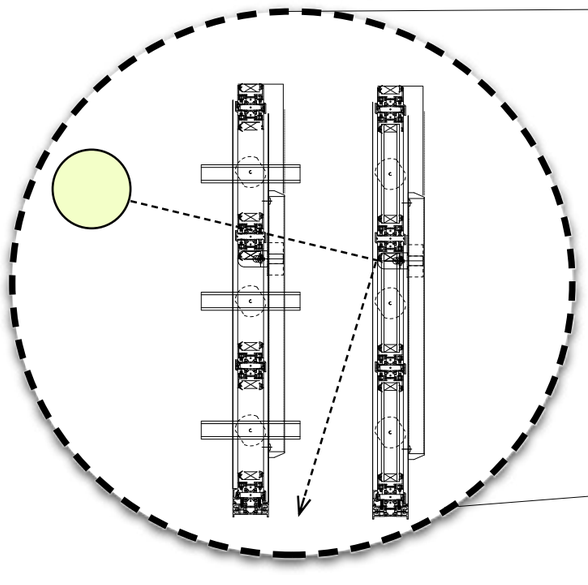


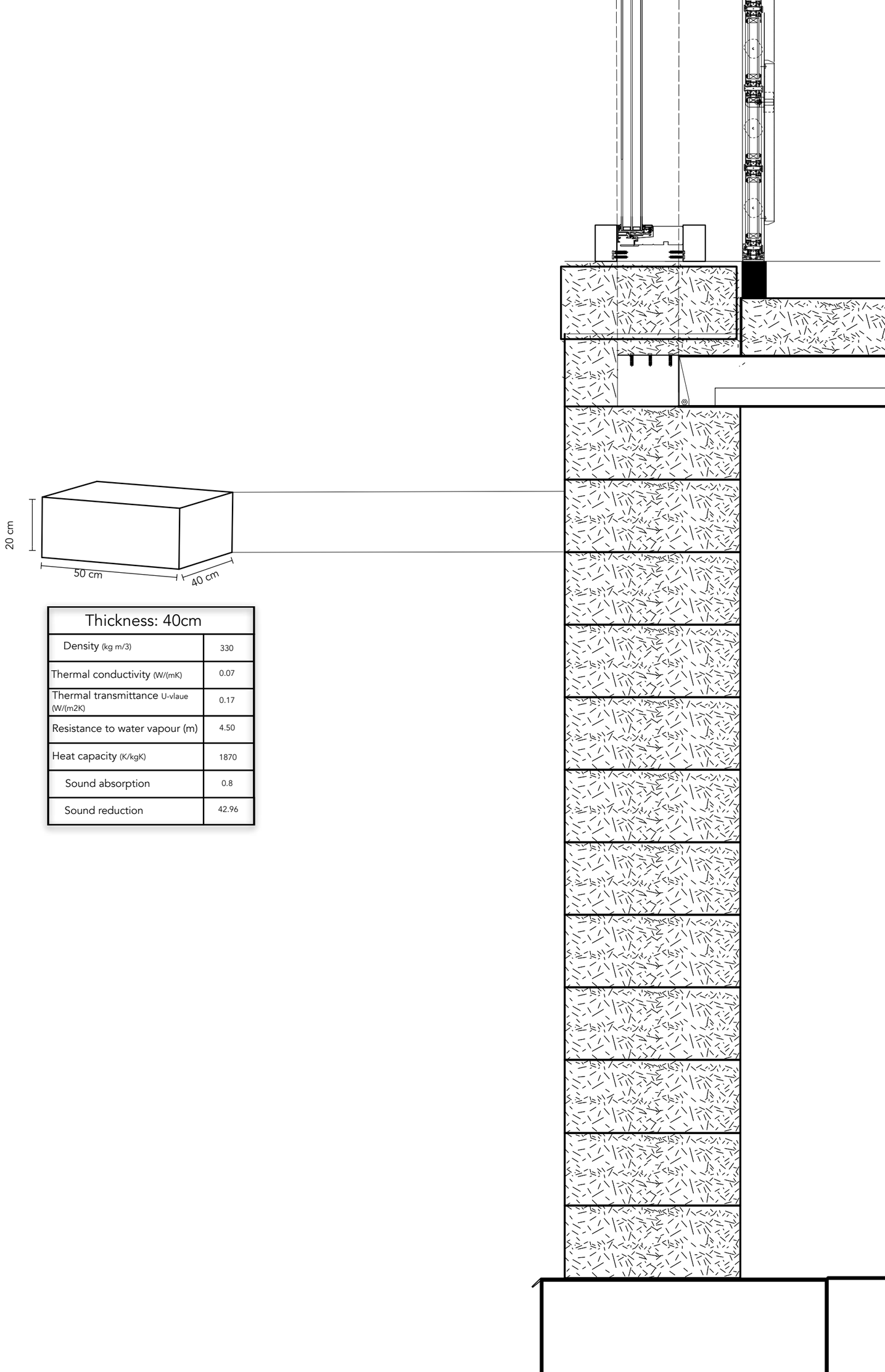






Performative testing: Detail section start of catalogue

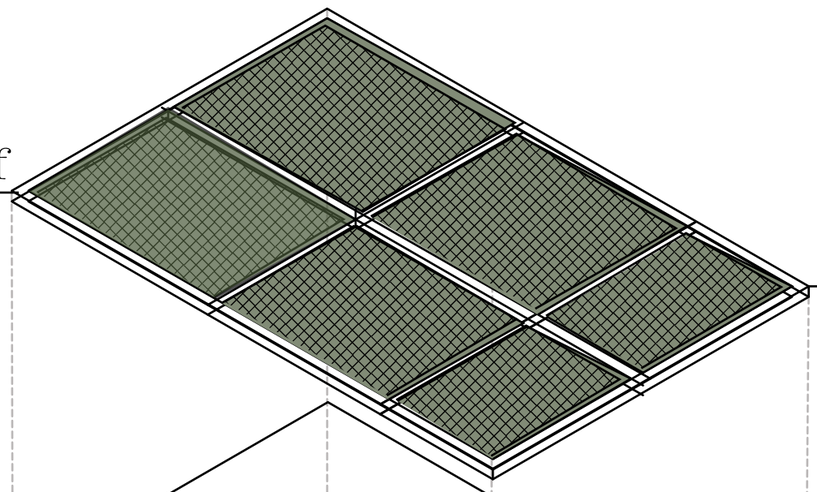




Thickness: 40cm	
Density (kg m/3)	330
Thermal conductivity (W/(mK)	0.07
Thermal transmittance U-vlaue (W/(m2K)	0.17
Resistance to water vapour (m)	4.50
Heat capacity (K/kgK)	1870
Sound absorption	0.8
Sound reduction	42.96

Green roof

The roof of the top building has a green roof to catch water and ensure the hempcrete is not damaged, it can be moved according to the building project.
.



Green roof

Roof / Floor slabs

The floor slabs are made into panels and are lifted onto the floors placed on the structure, acting as roof and floor slabs. 12 slabs in total.

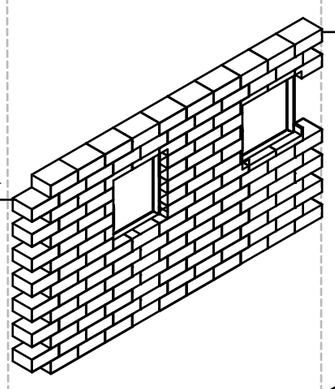
Floor slabs

1 .190cm x250cm
2 .298 x 250cm
3 .298 x 250cm

Wall Three

Back wall

The back of the house takes 140 bricks to construct- minus the bricks where the windows are placed. Both back windows are 1m by 1m.

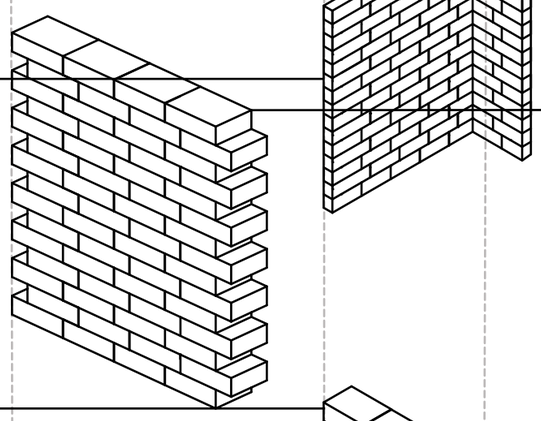


Internal Wall

280 bricks

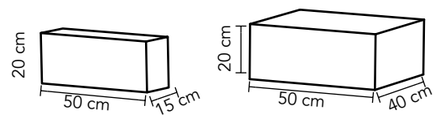
Internal Walls

The internal walls need no support and bricks can be layered and cut where doors are needed. 280 bricks are needed.



Wall Three

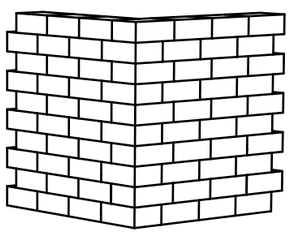
56 bricks

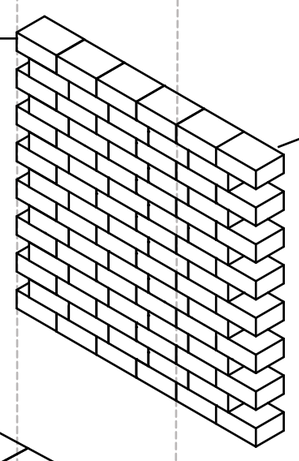


Internal / External

Walls

The wall is made from bricks using a stretcher bond there is no material waste and the only additions to the bricks would be to cut them to fit around the support pillars. The sides are the same so in total there are three walls but doubled up to make six in total for both sides. Taking 476 bricks to make both sides.





Wall Two

84 bricks

Wall One

98 bricks

Window

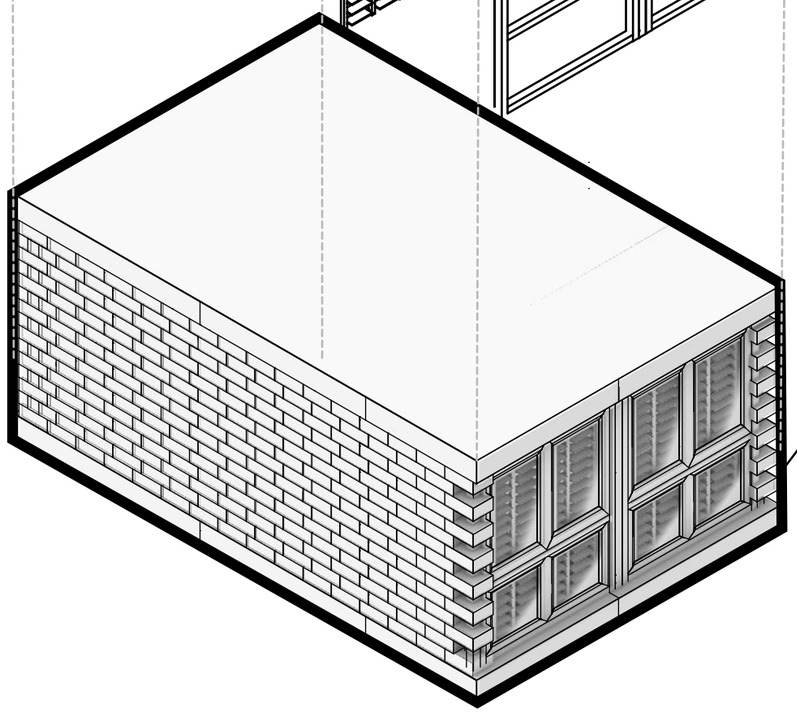
Louvers

Frame

Lintel
Pillar

Window

The front part consists of: frame sealed by hempcrete, windows, and louvers to block out the sun annually. The frame uses a lintel to support the window which is attached to the support pillars. Together it forms the front part of the home.

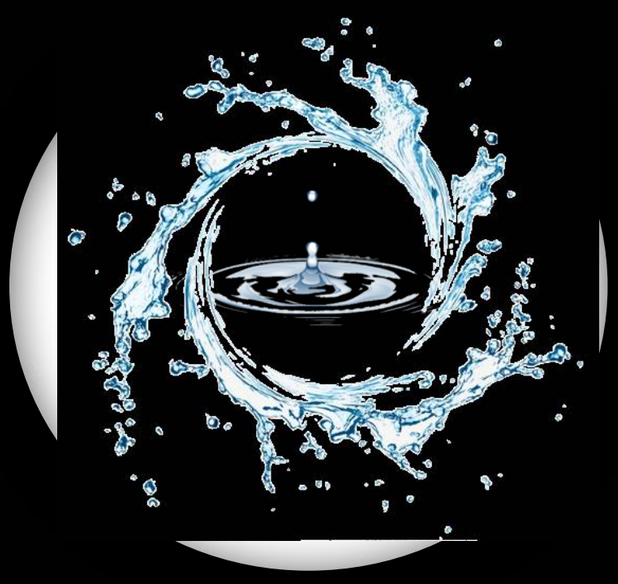


Housing

Assembly startegy and construction system: Assembly



1. Mix binder with water



Finish with the roof

Add the back and front
according to the catalogue

Finish and add lime render

Build strucrual frame

Contine to stack acroding to
brick formation

Add bricks between the fame

Add the internal bricks



2. Continue to mix while adding the hemp

